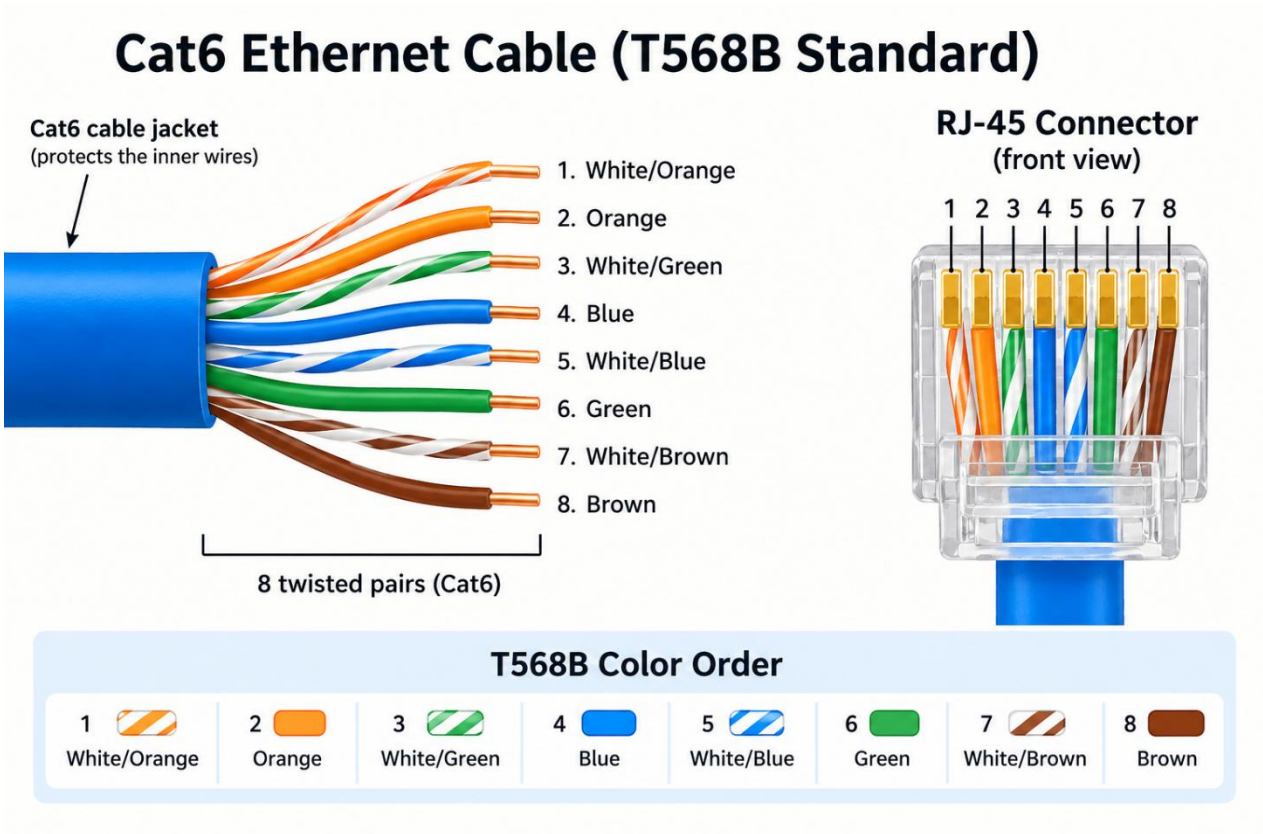
Assignment – 1.4

Network Cable and connectors

Q - 1. Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.



<u>Q - 2.</u> List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

Network Cable	Real-World Application / Device	Why It Is Used
1. Cat5e	Wi-Fi router	Provides a reliable wired connection with speeds up to 1 Gbps for internet access.
2. Cat6	Zomato POS terminal in a restaurant	Provides a fast, stable connection for processing orders and payments.
3. Fiber Optic	IPL live-streaming server	Supports very high data speeds over long distances, making it suitable for streaming live cricket matches.

Q - 3. Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

➢ Steps to Crimp an RJ-45 Connector:

1. **Cut the cable:** Cut the network cable to the required length.
2. **Remove the outer cover:** Use a wire stripper to remove a small part of the cable's outer covering.
3. **Arrange the wires:** Arrange the 8 wires in the T568B color order: white-orange, orange, white-green, blue, white-blue, green, white-brown, brown.
4. **Straighten the wires:** Straighten the wires and cut their ends evenly.
5. **Insert into RJ-45:** Insert all 8 wires into the RJ-45 connector in the correct order.
6. **Crimp the connector:** Use a crimping tool to press the connector firmly onto the cable.
7. **Test the cable:** Use a cable tester to check whether all wires are connected correctly.

➢ Two Common Mistakes to Avoid:

1. **Wrong wire order:** Always follow the T568B color sequence.
2. **Loose connection:** Make sure all wires reach the end of the connector and the cable is crimped firmly.

Q - 4. Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

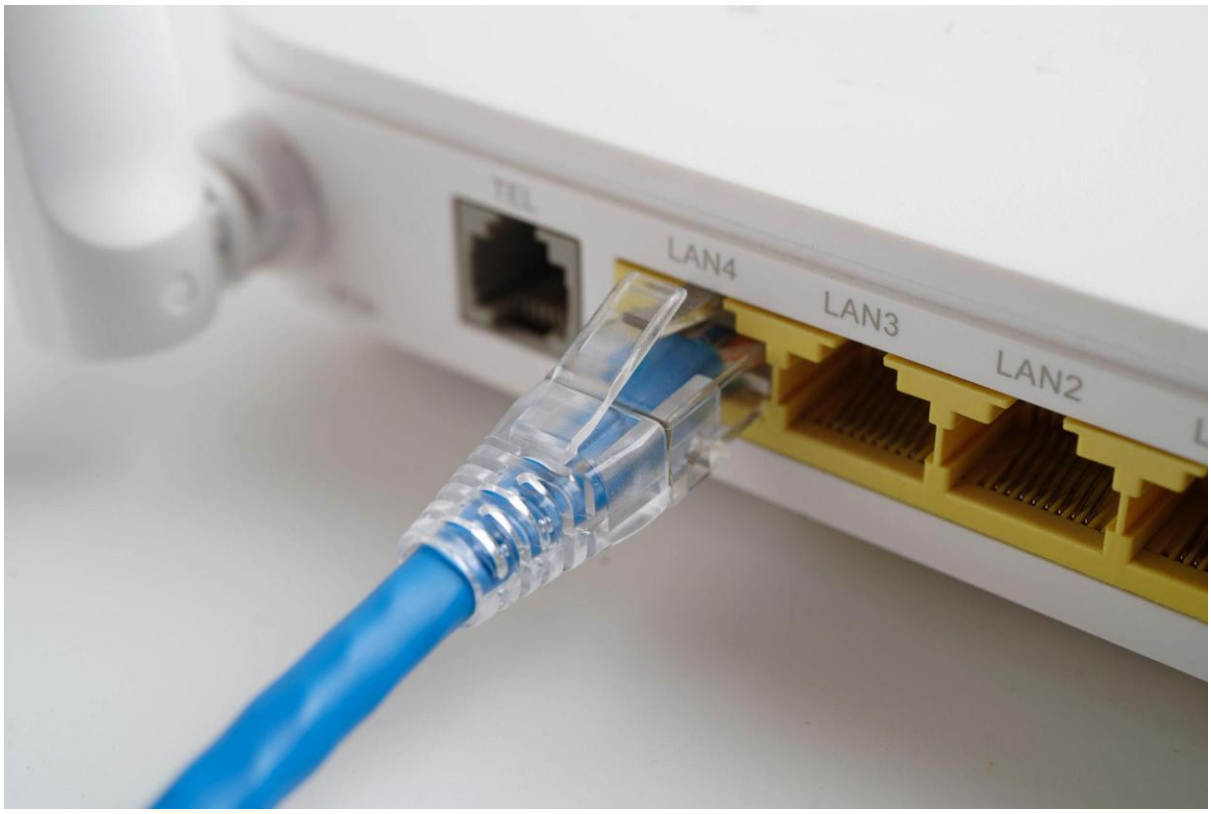
- **Cable Selection for an IPL Gaming Zone:**

1. **HDMI cable:** Used to connect gaming consoles and set-top boxes to TVs or projectors. It supports high-quality video and audio.

2. **Cat6 Ethernet cable with RJ-45 connector:** Used to connect gaming consoles, routers, and network switches. It provides fast and stable internet with low lag.

3. **Fiber optic cable:** Used for long-distance, high-speed connections between network switches. It supports fast data transfer with less signal loss.

<u>Q - 5.</u> Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects. Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.



- **Description:** An Ethernet cable (Cat6) has an RJ-45 connector at each end. It connects a Wi-Fi router to a computer, switch, or other network device. It

provides a fast and reliable wired internet connection.

- **Type of cable:** Ethernet LAN cable (RJ-45 connector).
- **Devices connected:** Wi-Fi router and computer or network switch.
- **Use:** It transfers data and provides a stable, high-speed network connection.